Hotel Booking & Reservation System Re-modeling

This presentation explains the problem, objectives, architecture, processes, data model, security, and expected outcomes for a digital hotel reservation platform.



IC 1202 - Information System Modeling

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Problem Statement(As we investigated An existing normal hotel booking system)

Manual, paper-based booking processes cause frequent errors, double-bookings, lost records, slow billing, and poor reporting. These issues lead to guest dissatisfaction, revenue leakage, and inefficient staff workflows.

Key loopholes

- Paper records → data loss and hard audits
- Double bookings and check-in delays
- Manual invoicing → billing mistakes

Business impact

- Lower guest satisfaction
- Lost revenue opportunities
- Inefficient staff time use





Project Objectives

Move from paper to a digital system that provides accurate, fast, and safe booking operations.

Real-time room tracking

Instant availability, holds, and status updates across channels.

Automated billing

Accurate invoices, tax calculations, discounts, and payments.

Efficient data retrieval

Fast searches for reservations, guest history, and audit logs.

System Context (ISM)

Who interacts with the system and the technology backbone that supports it.

Primary users

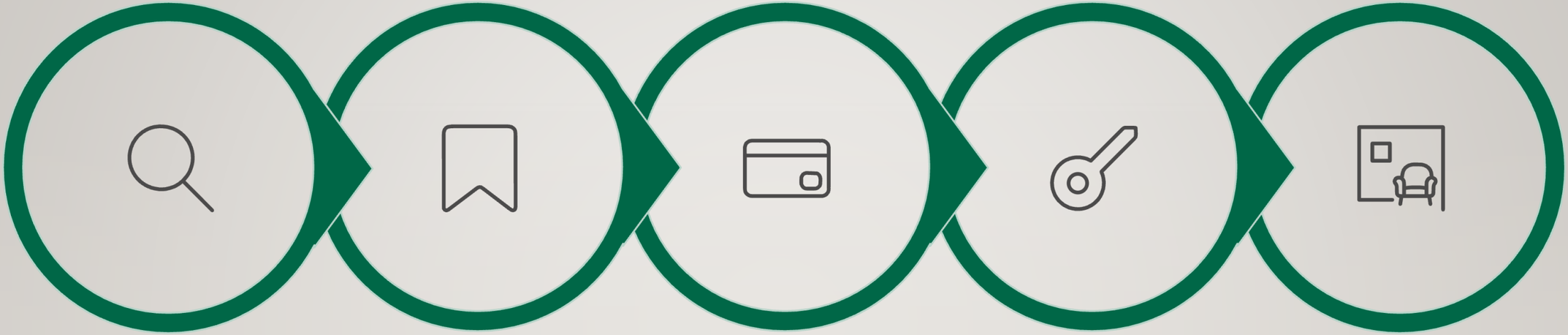
- Admin — system configuration, reports, user management
- Receptionist — bookings, check-in/out, room assignment
- Manager — revenue reports, occupancy dashboards
- Customer — search, book, pay, view invoices

Technology stack

- Frontend: React or Vue for responsive UI
- Backend: Node.js / Python (RESTful API)
- Database: PostgreSQL (ACID) with relational schema
- Payments: PCI-compliant gateway integration
- Hosting: Cloud (AWS / Azure)



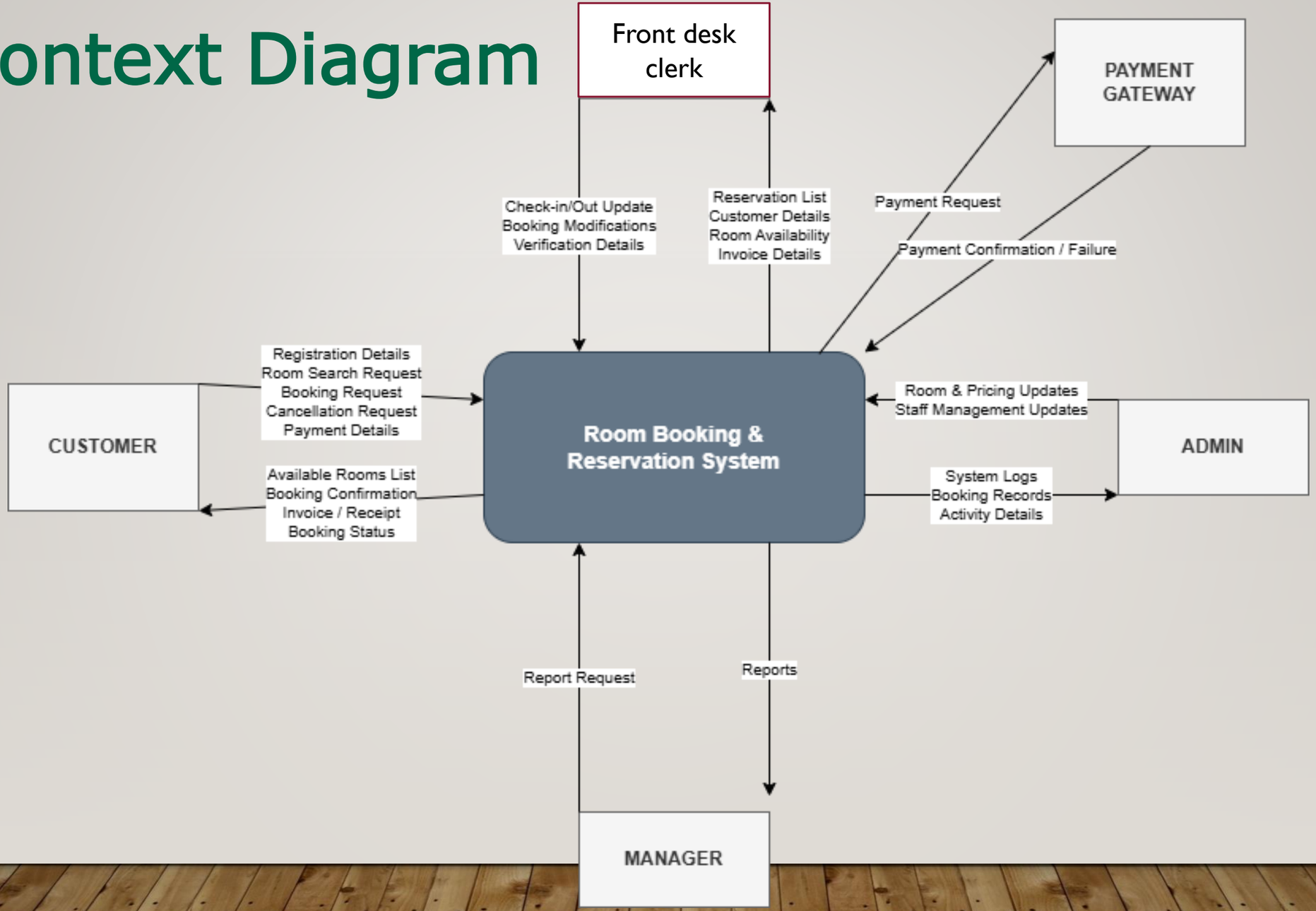
Business Process Flow



Search Rooms | **Select & Hold** | **Confirm & Authorize** | **Check-in & Assign** | **Manage Stay**

The diagram maps the customer journey and system actions: availability check, temporary holds to prevent double-booking, payment authorization at booking, live posting of charges during stay, and final invoice settlement at check-out.

Context Diagram



Operational Workflows & Module Responsibilities

Receptionist

Search availability, create/modify reservations, check-in/out, apply charges.

Manager

View occupancy dashboards, generate revenue reports, approve overrides.

Admin

Maintain room inventory, rates, tax settings, user accounts, and system configs.

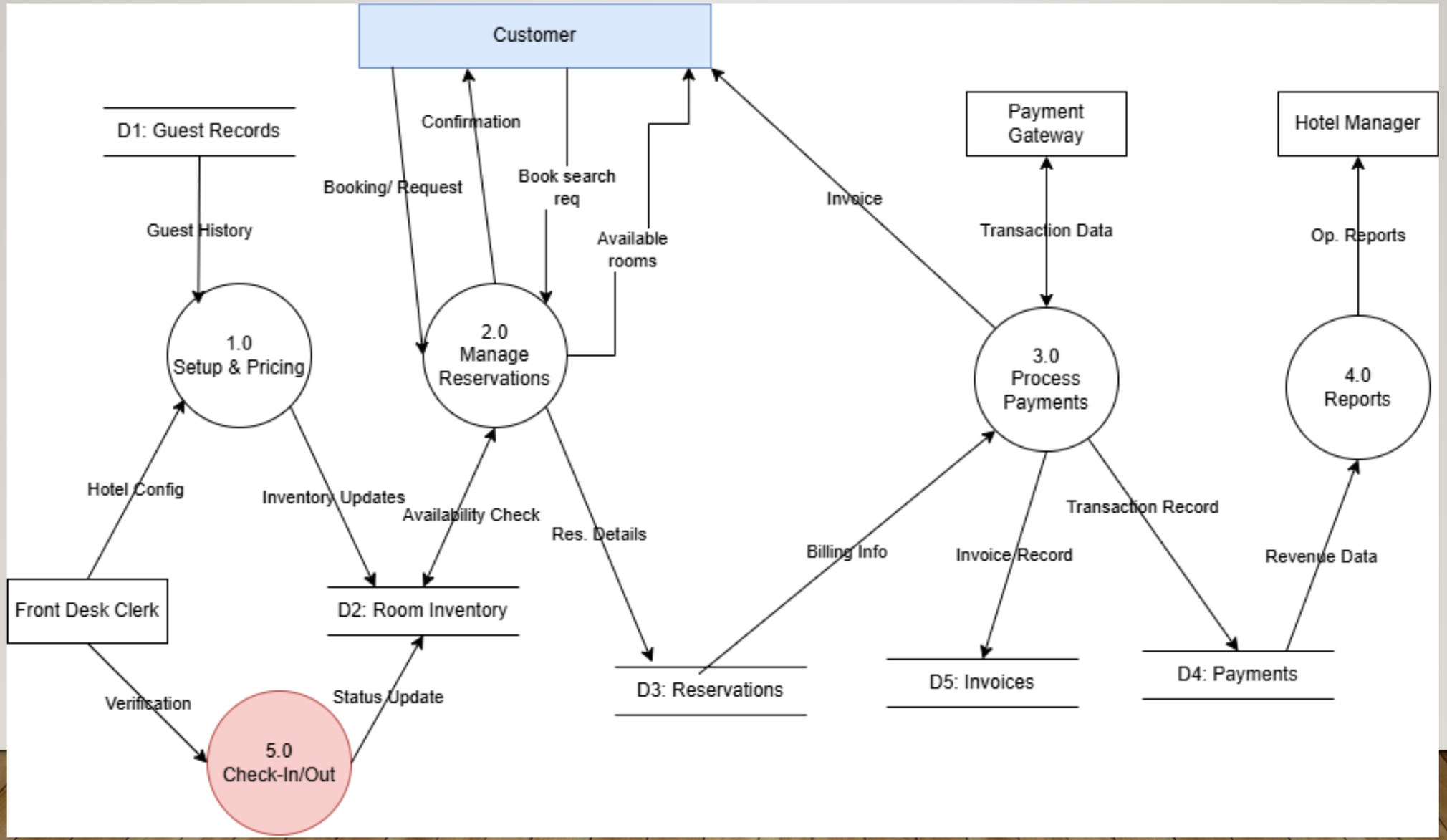
Customer

Self-service booking, payment, view/print invoices, cancel within policy.

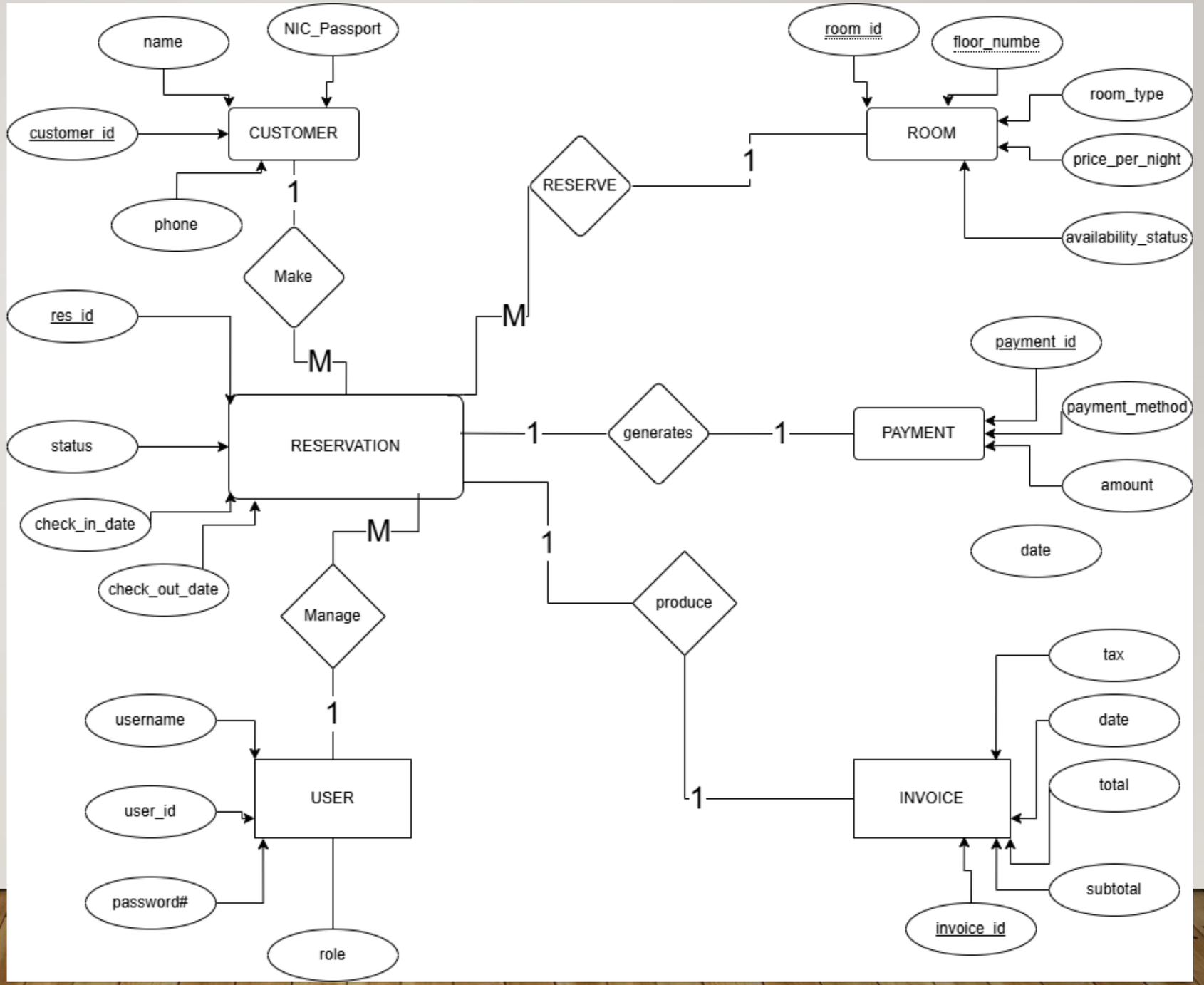
Example process: Receptionist places a temporary hold when a guest requests a room; system creates reservation and authorizes payment; hold converts to confirmed when payment completes; charges posted to reservation during stay; checkout triggers invoice generation.



DFD Diagram


Customer
D1: Guest Records
Confirmation
Booking/ Request
Book search req
Available rooms
Invoice
Payment Gateway
Transaction Data
Hotel Manager
Op. Reports
Guest History
1.0 Setup & Pricing
2.0 Manage Reservations
3.0 Process Payments
4.0 Reports
Hotel Config
Inventory Updates
Availability Check
Res. Details
Billing Info
Invoice Record
Transaction Record
Revenue Data
Front Desk Clerk
D2: Room Inventory
D3: Reservations
D5: Invoices
D4: Payments
Verification
Status Update
5.0 Check-In/Out

ER Diagram


name
NIC_Passport
customer_id
CUSTOMER
phone
1
Make
M
res_id
status
check_in_date
check_out_date
RESERVATION
M
Manage
1
username
user_id
password#
role
USER
M
RESERVE
1
room_id
floor_numbe
room_type
ROOM
price_per_night
availability_status
1
generates
1
payment_id
payment_method
amount
date
PAYMENT
1
produce
1
tax
date
total
subtotal
invoice_id
INVOICE

Database Design Details

Customer: Stores personal profiles including **customer_id (Primary Key), name, NIC / Passport (Unique), phone, and email (Unique) .**

Room: Tracks hotel inventory with **room_id (Primary Key), room_type, floor_number, price_per_night, and availability_status**

Reservation: Links customers to rooms using **res_id (Primary Key), customer_id (Foreign Key), room_id (Foreign Key), check_in_date, check_out_date, and booking status .**

Payment: Records financial transactions with **payment_id (Primary Key), res_id (Foreign Key), amount, payment_method, and payment_date**

Invoice: Automatically generated billing records **containing invoice_id, res_id, subtotal, tax, total, and generated_date.**

User (Staff): Manages internal access with **user_id, username, password_#, and defined role.**

Core tables and relationships enforce data integrity and allow strong queries for reporting and audits.

Entity Relationships Details

- Customer to Reservation (One-to-Many): A single customer can make multiple reservations .
- Room to Reservation (One-to-Many): One room can be reserved for different dates .
- Reservation to Payment (One-to-One): Each confirmed reservation produces one payment record .
- Reservation to Invoice (One-to-One): Every confirmed reservation generates a single invoice .
- User to Reservation (One-to-Many): A staff member (User) can handle and manage multiple bookings .

System modules and Security



Authentication

Username/password , Password policies



Role-based access

Manager, Receptionist, Guest. permissions per role.



Data backup & retention

Daily backups, weekly full snapshots



Payments & compliance

Use Safer gateway,

Expected Outcomes & Next Steps

Measured improvements

- ~90% reduction in booking errors (target)
- 100% billing accuracy with automated invoice generation
- Faster check-in/out and improved guest satisfaction

Implementation roadmap (next 90 days)

1. Design & data model finalization (2 weeks)
2. Core module development: bookings, payments, rooms (6 weeks)
3. Integration, testing, and staff training (4 weeks)

Success criteria

- Automated reports available daily
- Audit logs for all booking/payment actions
- Backup restore verified monthly

Questions and Discussions

We welcome Questions about our proposed system for Hotel booking and reservation
Please feel free to ask anything about the system's design, functionality, or implementation plan



Thank You for Your Time

We sincerely appreciate your valuable attention and insightful feedback during this presentation.

